

RYAN RAMDIAL

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Highlights of Qualification

- 5+ years experience in DWDM optical technologies for metro, long-haul, submarine and DCI systems
- Knowledgeable in optical network components such as transceivers, amplifiers (EDFA, Raman), multiplexers, fibers, etc.
- Experience with DSP, modulation formats (QAM, QPSK, etc.), and compensation algorithms
- Extensive use of Matlab programming for optical system modelling, designing and planning networks
- Automation and scripting experience using Python
- Canadian citizen, open to relocation to the US and Europe

Experience

R&D Optical Engineer
Ciena Corporation

July 2020 – Present

- Developing proprietary tools in Matlab for modelling performance of coherent modems
- Testing, verifying and optimizing optical networks for greenfield and brownfield scenarios
- Modelling penalties for link budgeting of system, such as (O)SNR margins (PMD, PDL, SPM, XPM, crosstalk, chromatic dispersion, etc.)
- Improving and testing link budgeting for both internal tools and external customers
- Scripting and automating new tools and processes to improve efficiency across team
- Supporting RFP/RFQ responses and aiding sales teams with new product capabilities

Research Assistant
Corning Inc.

September 2017 – December 2020

- Researched multi-subcarrier fiber-optic transmission systems
- Created and modified algorithms to optimize the digital signal processing at the coherent receiver
- Verified experimental results to ensure they coincided with theory

Education

Master of Applied Science, Electrical Engineering
McMaster University, Hamilton, ON

September 2018 – June 2020

- Researched the performance of multi-subcarrier fiber-optic communication systems
- Utilized Matlab to create and optimize algorithms for MSC signals

Bachelor of Engineering, Electrical Engineering
McMaster University, Hamilton, ON

September 2013 – May 2018

- Completed a 16-month internship prior to final year